

Apply filters to SQL queries

Project description

In this project, I used SQL to investigate potential security issues within an organization's systems. I applied filters using **AND**, **OR**, **NOT**, and **LIKE** operators to analyze login attempts and employee data. These queries helped identify suspicious login activity and determine which employee machines required security updates.

Retrieve after hours failed login attempts

```
SELECT *  
FROM log_in_attempts  
WHERE login_time > '18:00' AND success = '0';
```

Retrieve login attempts on specific dates

```
SELECT *  
FROM log_in_attempts  
WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
```

Retrieve login attempts outside of Mexico

```
SELECT *  
FROM log_in_attempts  
WHERE NOT country LIKE 'MEX%';
```

Retrieve employees in Marketing

```
SELECT *  
FROM employees  
WHERE department = 'Marketing' AND office LIKE 'East-%';
```

Retrieve employees in Finance or Sales

```
SELECT *  
FROM employees  
WHERE department = 'Finance' OR department = 'Sales';
```

Retrieve all employees not in IT

```
SELECT *  
FROM employees  
WHERE NOT department = 'Information Technology';
```

Summary

In this activity, I queried login attempt records to identify failed after-hours access, activity on specific dates, and login attempts occurring outside of Mexico. I also retrieved employee data based on department and office location to support targeted security updates. This project demonstrates how SQL filtering techniques can be used to support real-world cybersecurity investigations and operational decision-making.